



**SRI LANKA INSTITUTE OF ADVANCED TECHNOLOGICAL
EDUCATION**

Higher National Diploma in Information Technology

HNDIT 4052

Programming Individual Project

SOFTWARE REQUIREMENT SPECIFICATION

Hall Booking and Management System

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KUR/IT/2324/F/0057

ACADEMIC YEAR: 2324

YEAR II, SEMESTER I

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1. INTRODUCTION

1.1 Purpose of this Document

This Software Requirements Specification (SRS) document provides a detailed description of the **Home Bakers Product Marketplace System**. The purpose of this document is to clearly define the functional and non-functional requirements of the proposed web-based system.

This document serves as a formal reference between the client, developer, and academic supervisor by explaining the expected features, system behaviour, operational flow, constraints, and overall objectives of the platform. It also acts as a guideline during the planning, design, development, testing, deployment, and maintenance phases of the Software Development Life Cycle (SDLC).

The proposed system is designed to connect local home bakers and customers through a centralized online marketplace platform. The system enables bakers to upload and manage bakery products, maintain stock availability, and handle customer orders, while customers can browse products, place orders, track deliveries, and provide ratings and feedback. The platform also includes centralized inventory management, delivery validation, order tracking, payment handling, and administrative monitoring features to improve operational efficiency and user experience.

This document is prepared following standard software engineering practices and is intended for both academic and development purposes.

1.2 Scope of the System

The Home Bakers Product Marketplace System is a web-based application developed to automate and simplify the process of buying and selling homemade bakery products through a centralized online marketplace platform.

The system provides functionalities for:

- Real-time product browsing and product availability checking
- Customer and baker registration with secure login authentication
- Bakery product upload and management
- Inventory and stock management
- Online ordering and cart management
- Delivery validation based on minimum order value and delivery distance
- Order tracking and status updates
- Payment handling and payment status tracking
- Ratings and feedback management
- Admin, baker, and customer role management
- Sales, order, and inventory analytics

The system is specifically designed for local home bakers who need an online platform to promote and manage their bakery products efficiently while providing customers with a convenient and user-friendly ordering experience.

The project focuses on improving operational efficiency, reducing manual workload, managing inventory effectively, simplifying delivery and order processing, and enhancing customer experience through digital transformation.

1.3 Definitions and Acronyms

Acronym	Meaning
SRS	Software Requirements Specification
UI	User Interface

API	Application Programming Interface
REST	Representational State Transfer
CRUD	Create, Read, Update, Delete
OTP	One Time Password
HTTPS	Hypertext Transfer Protocol Secure
DB	Database
SDLC	Software Development Life Cycle
SQL	Structured Query Language
MySQL	Relational Database Management System used for storing data in tables
React	JavaScript library used for building user interfaces
Node.js	JavaScript runtime used for backend development
Admin	System administrator who manages the platform
Inventory	Stock management system for products
Order Tracking	Process of monitoring order status from placement to delivery
Authentication	Process of verifying user identity
Authorization	Process of controlling user access levels
Dashboard	Central control panel for system monitoring and management
Delivery Validation	Process of checking delivery distance, stock availability, and order conditions before confirming order

1.4 System Overview

The Home Bakers Product Marketplace System is designed as a centralized digital platform that replaces traditional product selling and order management methods such as social media messaging, phone calls, and manual order tracking.

The system enables customers to browse bakery products, search and filter items, view detailed product information, place online orders, track delivery statuses, and provide ratings and feedback through an interactive and user-friendly web interface. Bakers can upload and manage bakery products, monitor stock and inventory levels, handle customer orders, and configure delivery constraints such as minimum order value and delivery distance limitations. Administrators can manage users, monitor marketplace activities, track orders, payments, and deliveries, manage inventory operations, and generate analytical reports for decision-making.

The platform incorporates centralized stock and delivery management, where the system automatically tracks product availability, validates delivery conditions, updates order statuses, and ensures efficient order processing. It also supports payment handling through Cash on Delivery, with the possibility of integrating online payment gateways in future enhancements.

The application is developed using modern web technologies including React.js for the frontend, Node.js for the backend, and MySQL for database management, ensuring a responsive, secure, scalable, and efficient marketplace environment.

2. GENERAL DESCRIPTION

2.1 Product Perspective

The Home Bakers Product Marketplace System is a standalone web-based marketplace application that integrates product management, inventory tracking, online ordering, delivery validation, payment handling, order tracking, customer feedback, and administrative monitoring into one centralized system.

The system replaces traditional product selling methods such as social media messaging, phonebased ordering, and manual stock tracking with a modern digital solution that improves product management, operational efficiency, inventory control, and customer experience.

The system consists of three major user groups:

Admin

The administrator has full access to the system. Admin users can manage customers, bakers, products, inventory, orders, delivery operations, payments, analytics, reports, and overall marketplace activities.

Baker

Bakers can register and manage their bakery products, upload product images, update stock availability, manage customer orders, configure delivery settings such as minimum order values and delivery distance limits, and monitor sales activities through a dedicated dashboard.

Customer

Customers can browse available bakery products, search and filter products, view product details, add items to cart, place online orders, make payments, track order statuses, and provide ratings and feedback through a user-friendly interface.

2.2 Product Functions

The major functions of the system are described below:

1. User Authentication

The system provides secure login and registration facilities for administrators, bakers, and customers using Firebase Authentication with role-based access control.

2. Product Management

Bakers can add, update, delete, and manage bakery product information including product name, description, price, images, stock quantity, and availability status.

3. Inventory Management

The system automatically tracks product stock quantities and updates inventory after successful orders to prevent unavailable product purchases.

4. Order Management

Customers can add products to cart and place online orders while the system validates stock availability, minimum order value, and delivery distance before confirming the order.

5. Delivery Management

The system manages delivery validation and order tracking by checking delivery eligibility, updating order statuses, and monitoring delivery progress from order confirmation to final delivery.

6. Payment Management

The system supports payment tracking for customer orders using Cash on Delivery and optional online payment integration features.

7. Ratings and Feedback System

Customers can provide ratings and feedback for bakery products and services after successful order completion.

8. Notification System

The system provides notifications and status updates related to orders, deliveries, payments, and inventory activities through system alerts.

9. Reporting and Analytics System

Administrators can generate reports and analytics including sales reports, order statistics, revenue summaries, inventory reports, and user activity monitoring.

2.3 User Classes and Characteristics

2.3.1 Administrator

The administrator is responsible for managing and monitoring the entire marketplace system. The admin user is expected to have basic technical knowledge and understanding of marketplace, inventory, and order management operations.

Admin users perform activities such as:

- Managing bakers and customers
- Monitoring orders, payments, and deliveries
- Managing inventory and stock activities
- Generating reports and analytics
- Monitoring marketplace operations
- Updating order statuses and system settings

2.3.2 Baker

Bakers are users who sell homemade bakery products through the platform. Bakers are expected to have basic computer and internet literacy to manage products and customer orders effectively.

Bakers can:

- Register and login
- Upload and manage bakery products
- Update stock quantities and product availability
- Set minimum order values and delivery distance limits
- View and manage customer orders
- Accept or reject orders
- Monitor sales and revenue summaries

2.3.3 Customer

Customers are users who browse and purchase bakery products through the platform. Customers are expected to have basic computer and internet literacy.

Customers can:

- Register and login
- Browse and search bakery products
- View product details and ratings
- Add products to cart
- Place online orders
- Make payments
- Track order statuses
- Provide ratings and feedback
- Receive notifications and order updates

2.4 Features and Benefits

The proposed system provides several important benefits:

- Centralized digital marketplace management
- Real-time product browsing and stock availability checking
- Automated inventory and stock management
- Prevention of unavailable product orders through stock validation
- Online order placement and order tracking
- Delivery validation based on minimum order value and delivery distance
- Secure and organized user, product, and order management
- Role-based access control for Admin, Baker, and Customer users
- Automated order status updates and notifications
- Payment tracking and management
- Ratings and feedback management for improved customer trust
- Dynamic sales, revenue, inventory, and order reports
- Improved customer satisfaction and operational efficiency
- Increased online visibility and business opportunities for local home bakers

2.5 Assumptions and Dependencies

The system assumes that:

- Users have stable internet connectivity
 - MySQL database services remain operational and accessible
 - Bakers regularly update product details, stock quantities, and delivery settings
 - Customers provide accurate delivery addresses and order information
 - Delivery operations are available within the configured delivery distance limits
 - Users have basic computer and internet literacy to use the platform effectively
- The system

depends on:

- Firebase Authentication
- MySQL Database
- React.js frontend framework
- Node.js backend runtime environment
- Internet connectivity for real-time operations
- Cloud hosting platform for deployment and storage services

3. FUNCTIONAL REQUIREMENTS

3.1 User Authentication

The system shall provide secure user authentication features for all users (Admin, Baker, and Customer).

Functional Requirements:

- Users shall be able to register using email and password
- Users shall be able to log in securely using authentication mechanisms (JWT-based authentication or Firebase Authentication)
- The system shall support role-based access control (Admin, Baker, Customer)
- Invalid login attempts shall display appropriate error messages
- Passwords shall be securely hashed and stored in the database to ensure data protection

- The system shall maintain session management for authenticated users

3.2 Hall Management

The baker shall manage bakery product information through the baker dashboard, while the administrator has full control over all products in the system.

Functional Requirements:

- Add new bakery products
- Update product information such as name, description, and price
- Delete existing products
- Manage product stock (inventory quantity)
- Upload product images and descriptions
- Set product availability status (In Stock / Out of Stock)
- View product listing and availability status

3.3 Booking Management

The system shall provide order management functionalities for both customers, bakers, and administrators.

Functional Requirements:

- Customers shall be able to place orders through the cart and checkout system
- The system shall validate product stock availability before confirming orders
- The system shall validate delivery conditions such as minimum order value and delivery distance
- The system shall prevent ordering of out-of-stock products
- Order statuses shall include: o Pending o Accepted o Preparing o Out for Delivery o Delivered o Cancelled
- Bakers shall be able to accept or reject incoming orders
- Administrators shall be able to monitor, manage, and update all order statuses
- The system shall maintain order history for customers and bakers

3.4 Package and Service Management

The system shall support centralized inventory control and delivery-based validation for all bakery products.

Functional Requirements:

- Bakers shall be able to manage product stock quantities
- The system shall automatically update stock after each successful order
- The system shall prevent ordering of products that are out of stock
- Administrators shall be able to monitor overall inventory levels across all bakers
- The system shall support minimum order value settings per baker
- The system shall support delivery distance limitations based on baker location
- The system shall validate delivery eligibility before confirming orders
- The system shall display stock availability in real time to customers
- The system shall maintain inventory logs for tracking stock changes

3.5 Payment Management

The system shall provide secure payment handling and payment tracking functionality for customer orders.

Functional Requirements:

- Customers shall be able to make payments using Cash on Delivery and/or online payment methods (optional integration)
- The system shall support payment status tracking (Pending, Paid, Failed)
- The system shall automatically update payment status after successful transactions
- The system shall calculate and display total payable amount during checkout
- The system shall maintain complete payment history for each customer
- The system shall generate and store digital receipts or transaction records securely
- The system shall ensure secure handling of payment-related data
- Sandbox or test payment gateways may be used for demonstration purposes

3.6 Notification System

The system shall provide automated notifications to improve communication between customers, bakers, and administrators.

Functional Requirements:

- The system shall send order confirmation notifications after a successful order is placed
- The system shall send order status updates (Pending, Accepted, Preparing, Out for Delivery, Delivered)
- The system shall send payment status notifications (Payment Pending, Paid, Failed)
- The system shall send stock alerts when products are low or out of stock
- The system shall notify bakers when new orders are received
- The system shall support system alerts and in-app notifications
- The system shall allow users to view and manage notifications (mark as read / delete)
- The system may support email notifications for important updates

3.7 Reporting and Analytics

The administrator shall have access to dynamic reports and analytics to monitor marketplace performance and decision-making.

Functional Requirements:

- The system shall generate daily, weekly, and monthly sales reports
- The system shall provide order summary reports based on time period
- The system shall generate payment reports including pending, completed, and failed transactions
- The system shall provide inventory reports showing stock levels and product availability
- The system shall display most ordered products and best-performing bakers
- The system shall show customer activity and order history analytics
- The system shall provide graphical charts for sales trends and order statistics
- The system shall support a dynamic admin dashboard with real-time analytics

3.8 User Role Management

The system shall implement role-based access control (RBAC) to ensure secure and restricted access to system functionalities.

Functional Requirements:

- The system shall define three user roles: Admin, Baker, and Customer
- Admin users shall have full access to all system modules and data
- Bakers shall have access only to product management, order handling, and inventory-related features
- Customers shall have access only to product browsing, ordering, payment, and feedback features
- The system shall restrict unauthorized access to admin and baker functionalities
- The system shall enforce role-based access control during authentication and navigation
- The system shall ensure secure session handling for each user role
- The system shall validate user permissions before allowing access to sensitive operations

4. INTERFACE REQUIREMENTS

4.1 User Interface

The system shall provide a modern, responsive, and user-friendly interface for all users (Admin, Baker, and Customer).

User Interface Features:

- Responsive design compatible with desktop, tablet, and mobile devices
- Clean and modern marketplace-themed UI design suitable for bakery products
- Separate dashboards for Admin, Baker, and Customer roles
- Interactive product browsing and filtering interface
- Responsive product card layouts for easy viewing
- Shopping cart and checkout interface with clear order summary
- Toast notifications and alert messages for user actions
- Intuitive navigation menu with role-based access menus
- Dashboard charts and analytics UI for admin and baker users

4.2 Software Interface

The system integrates several modern technologies and third-party services to ensure smooth functionality, scalability, and secure operation.

Technology	Purpose
React.js	Frontend development for user interface
Node.js + Express.js	Backend API development and business logic handling
MySQL	Database management for storing system data
Firebase Authentication	Secure user authentication and role management
Cloud Storage (e.g., Firebase Storage / AWS S3)	Image and file storage for product images
Re charts	Data visualization and analytics dashboard
REST API systems	Communication between frontend and backend

4.3 Communication Interface

The system uses secure and standardized communication protocols to ensure reliable data exchange between frontend, backend, and external services.

Communication Requirements:

- The system shall use HTTPS protocol for secure communication between client and server
- The system shall use JSON format for all REST API request and response data exchange
- The system shall implement RESTful APIs for communication between React frontend and Node.js backend
- The system shall ensure secure authentication using tokens (JWT-based authentication or Firebase Auth tokens)
- The system shall support real-time updates for order status, inventory changes, and notifications
- The system shall ensure secure data transmission to prevent unauthorized access and data leakage

5. DESIGN CONSTRAINTS

The following design constraints apply to the system:

- The application must be a web-based platform only
- The system must be developed using React.js for frontend development
- The system must use Node.js and MySQL for backend and database operations
- The system must support modern web browsers such as Chrome, Firefox, and Edge
- The system shall follow responsive web design principles to support mobile, tablet, and desktop devices
- GitHub shall be used for version control and collaborative development tracking
- Secure coding practices shall be followed to protect user data and system integrity
- The system shall be suitable for small and medium-scale home bakery businesses
- The system shall rely on internet connectivity for all real-time operations (orders, stock, and notifications)

6. NON-FUNCTIONAL REQUIREMENTS

6.1 Performance

- System response time shall remain below 3 seconds for most user requests
- Product listing and search results shall load within 2 seconds
- The system shall support multiple concurrent users (customers, bakers, and admin)
- Order processing and checkout operations shall be completed without significant delay

6.2 Security

- User passwords shall be securely hashed and stored in the database
- Role-based access control shall be enforced for Admin, Baker, and Customer roles
- The system shall protect sensitive data using secure authentication mechanisms (JWT/Firebase Auth)
- The system shall prevent unauthorized access and common web attacks such as SQL injection and XSS
- Secure HTTPS communication shall be used for all data transmission

6.3 Reliability

- The system shall ensure data consistency across all transactions (orders, inventory, and payments)
- Order and payment records shall remain accurate and permanently stored
- The system shall provide backup support for database recovery in case of failure
- The system shall ensure high availability during peak usage

6.4 Usability

- The system shall provide a simple, intuitive, and user-friendly interface
- Clear validation messages and error notifications shall be displayed to users
- The system shall be easy to use for non-technical users such as home bakers and customers
- Navigation shall be consistent across all pages and user roles

6.5 Scalability

- The system architecture shall support future expansion of users and products

- Multiple bakers and large product catalogs shall be supported without performance degradation
- The system shall allow future integration of mobile applications and advanced features
- Database structure shall support horizontal growth and additional modules

6.6 Maintainability

- The system shall follow modular code architecture for easy maintenance
- Frontend and backend components shall be reusable and well-structured
- Code shall be properly documented for future development and debugging
- Future updates and feature enhancements shall be easy to implement

7. APPENDICES

7.1 Assumptions

- Users have stable internet connectivity
- Bakers regularly update product, stock, and order information
- The system database and server remain operational
- Users provide accurate delivery addresses and contact details
- External services (authentication and hosting) remain available and stable

7.2 Dependencies

- React.js Framework for frontend development
- Node.js + Express.js for backend development
- MySQL Database for data storage and management
- JWT / Firebase Authentication for secure login system
- Cloud hosting platform for deployment
- GitHub for version control and collaboration
- Recharts for analytics and data visualization

7.3 Acronyms

Acronym	Meaning
API	Application Programming Interface
UI	User Interface
DB	Database
SRS	Software Requirements Specification
HTTPS	Hypertext Transfer Protocol Secure
CRUD	Create, Read, Update, Delete
JWT	JSON Web Token
SQL	Structured Query Language
MVC	Model View Controller
SDLC	Software Development Life Cycle

7.4 References

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8. Use Case Diagram



8. Entity Relationship Diagram



